

Take-Away Games

Traps, tricks, and testing strategies

Name: _____

Partner: _____

How to play

- Make a pile of counters. Two players take turns.
- On your turn you **must** take some counters. The **rule** says how many you may take. You can't skip, and you can't take more counters than the pile has.
- Whoever takes the **last counter wins**.

Group of three? Two play and the third is the coach who checks every move. Switch each game.

Warm-up rule: take 1 or 2.

- 1 Play three games. Start with 10 counters each time.

Game	Who went first?	Who won?
1		
2		
3		

Win piles and trap piles

Imagine it is **your turn**. The pile is a **win pile (W)** if you have a way to win, no matter what your partner does. It is a **trap (T)** if you will lose whenever your partner plays well.

0 is a trap: if it is your turn and no counters are left, your partner just took the last one!

- 2 With the rule **take 1 or 2**, write W or T under each pile.

0	1	2	3	4	5	6	7	8	9	10	11	12
T												

- 3 What pattern do the traps make? _____

- 4 A game starts with 20 counters. Do you want to go first or second?

If you go first, how many will you take? Why? _____

The main game: take 1, 3, or 4



take 1



take 2



take 3



take 4

**Taking 2
is NOT
allowed!**

Last counter still wins. Careful: if only 2 counters are left, the only move is to take 1!

5 Play at least four games, starting with 10 counters each time.

Game	Who went first?	Who won?
1		
2		
3		
4		

6 Think back to the ends of your games. Was a player ever stuck with a pile they did not want? Which piles were bad to have on your turn?

7 Make a guess: with 10 counters, is it better to go first or second?

You will check your guess on the next page.

8 Play tiny games that start with 1, 2, 3, 4, or 5 counters. Try every first move! Circle the tiny piles that are traps for the player whose turn it is.

1

2

3

4

5

Pick one trap you found. Explain why every move from it is bad.

Trap finder for take 1, 3, or 4

How to find traps

- 0 is a trap.
- If **some** move leaves your partner a trap, the pile is a **W**.
- If **every** move leaves your partner a W, the pile is a trap, **T**.

Work from the top down. Each row shows how many counters are left after each move. **Circle** the leftover piles that are traps. Circled one? Write W. None circled? Write T.

pile	take 1	take 3	take 4	W or T
0	–	–	–	T
1	0	–	–	W
2	1	–	–	T
3	2	0	–	
4	3	1	0	
5	4	2	1	
6	5	3	2	
7	6	4	3	
8	7	5	4	
9	8	6	5	
10	9	7	6	

pile	take 1	take 3	take 4	W or T
11	10	8	7	
12	11	9	8	
13	12	10	9	
14	13	11	10	
15	14	12	11	
16	15	13	12	
17	16	14	13	
18	17	15	14	
19	18	16	15	
20	19	17	16	

9 Check your guess from page 2. Is 10 a W pile or a trap?

10 If you start with 10 and go first, how many should you take?

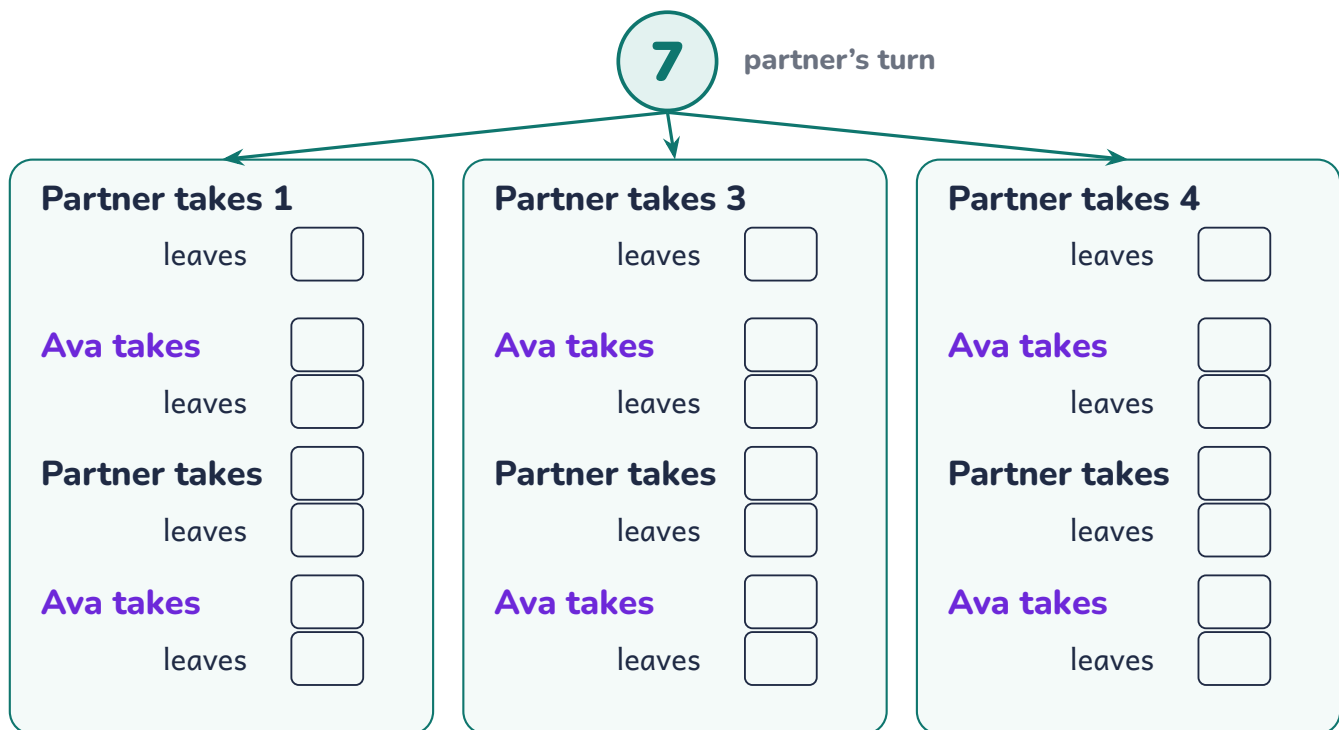
Check: from 1 to 20 you should find exactly 5 traps.

Test the claims

What makes a strategy work?

A **winning strategy** must have a good answer to **every** move your partner might make, not just the moves you expect. All the claims on this page use the rule **take 1, 3, or 4**.

- 11 Ava says:** “If there are 7 counters and my partner goes first, I can always win!” Check every first move her partner could make. Fill in each path until the pile is 0.



Is Ava right? How do you know? _____

- 12 Ben says:** “With 9 counters, I go first and take 1. That leaves 8, and 8 is a lot! So I will win.”

Find a move for Ben's partner that beats Ben's plan.

Partner takes , which leaves for Ben.

Why is that bad for Ben? _____

- 13 Cleo says:** “Always take 4. The biggest move is always the best move!”

Find a pile where taking 4 loses, but a different move wins.

Pile: Taking 4 leaves A better move is to take

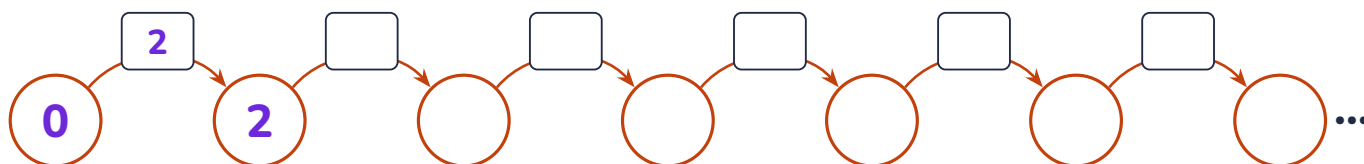
Patterns and predictions

Still using the rule **take 1, 3, or 4**.

14 Circle every trap from your trap finder in this chart. Then look for a pattern and keep circling all the way to 39.

0	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29
30	31	32	33	34	35	36	37	38	39

15 Write the traps in order. In each small box, write the jump from one trap to the next.



What do you notice about the jumps? _____

16 Use your pattern. Is 50 a trap?

How do you know? _____

17 A game starts with 25 counters and you go first. What should you take?

Explain how you know: _____

Challenge: invent a rule

Make up a new rule, for example **take 1, 2, or 3** or **take 1 or 4**. Play it, then find its traps.

My rule: take _____

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

What pattern do the traps make for your rule? _____